Image Captioning with Retrieval-Augmented Generation

Phase 1 Technical Report

Dataset Preparation, Embeddings, and Knowledge-Base Construction

Project repository:	https://github.com/djidji09/image-captioning-rag
Reviewed branch:	`ahmed-branch`
Dataset:	MS-COCO 2017
Execution environment:	Google Colab and Google Drive
Report date:	31 October 2026

Prepared for project review and Phase 1 assessment

Contents

1 Executive Summary **3**

2 Scope and Source Material Reviewed **3**

2.1 Materials reviewed 3

2.2 Evidence classification 4

2.3 Review limitations 4

3 Workspace Setup and Project Organization **4**

3.1 Execution environment 5

4 Dataset Audit, Scaling, and Knowledge-Base Rescaling **5**

4.1 Source dataset 5

4.2 Incomplete local extraction and restoration 6

4.3 Development sample versus final scale 6

5 End-to-End Pipeline and Implementation **7**

5.1 Workspace initialization 7

5.2 Dataset audit 7

5.3 Split construction 7

5.4 Preprocessing 8

5.5 Embedding generation 8

5.6 FAISS index construction 9

6 Drive Structure and Generated Artifacts **9**

7 Statistical Exploration and Dataset Results **10**

7.1 Caption inventory 10

7.2 Object statistics 10

7.3 Caption language statistics 11

7.4 Limits of the statistical analysis 11

8 Requirement-by-Requirement Evaluation **11**

9 Overall Re-evaluation, Risks, and Recommendations **13**

9.1 Conflicting split implementations 13

9.2 Test-set size inconsistency in the specification 13

9.3 Five-caption policy 14

9.4 Leakage and self-retrieval 14
9.5 Final-scale quality-control gap 14
9.6 Recommended actions before Phase 1 approval 14

10 Evidence Appendix **15**
10.1 Key numerical evidence 15
10.2 Model and index configuration 16
10.3 Final reported artifact paths 16
10.4 Important interpretation notes 16

11 Conclusion **17**

1 Executive Summary

Phase 1 focused on preparing a reproducible image-captioning knowledge base from the MS-COCO 2017 dataset. The work covered workspace preparation, dataset auditing, split construction, preprocessing checks, embedding generation, FAISS index construction, and quality-control activities.

The final-scale embedding and indexing notebook reports successful processing of 10,000 knowledge-base images and 50,030 associated captions. The image embeddings have shape $10{,}000 \times 512$, while the caption embeddings have shape $50{,}030 \times 384$. The corresponding FAISS indexes report 10,000 image vectors and 50,030 caption vectors.

The work also encountered an incomplete local extraction of the COCO training images. A total of 4,908 selected images were initially unavailable in the local training-image directory. A targeted restoration procedure subsequently reported 4,908 restored images and zero failures, making all 10,000 selected knowledge-base images accessible to the pipeline.

However, Phase 1 cannot yet be considered fully frozen or formally approved. The review identified conflicting split implementations, a specification inconsistency concerning the test-set size, and a mismatch between the final-scale embedding outputs and the development-scale quality-control outputs. In particular, the quality-control notebook still reports 1,000 images and 5,001 captions, rather than validating the final 10,000-image release.

Overall conclusion: the core final-scale knowledge-base construction and embedding pipeline have been demonstrated successfully, but formal Phase 1 completion and release approval remain conditional on resolving the split and specification conflicts and rerunning quality control against the final release artifacts.

2 Scope and Source Material Reviewed

This report evaluates the Phase 1 implementation using the available project materials and execution evidence.

2.1 Materials reviewed

The review covered the following repository materials on branch `ahmed-branch`:

- `README.md`
- `configs/phase1.yaml`
- `notebooks/00_workspace_setup.ipynb`
- `notebooks/01_dataset_audit.ipynb`
- `notebooks/02_split_and_statistics.ipynb`
- `notebooks/03_preprocessing.ipynb`
- `notebooks/04_embeddings_and_index.ipynb`

- `notebooks/05_quality_control.ipynb`
- `notebooks/06_final_split.ipynb`

The assessment also uses execution outputs and project details recorded during the development and debugging process, including the split manifest, restored-image counts, embedding dimensions, index sizes, and quality-control outputs.

2.2 Evidence classification

The following terms are used throughout the report:

- **Verified:** directly supported by notebook output, repository content, or an explicit execution result available during the review.
- **Partially verified:** some implementation or output is present, but the full requirement has not been demonstrated at the final release scale.
- **Not completed:** the required work is absent or the available evidence shows that it has not been performed.
- **Unable to verify:** the relevant artifact or execution evidence was not directly accessible for independent inspection.

2.3 Review limitations

The repository notebooks and their recorded outputs were reviewed, but the complete contents of the Google Drive release directory and all binary artifacts were not independently inspected. Consequently, notebook output and recorded artifact paths are treated as evidence that files were written, not as independent verification of every file's current existence, integrity, or reproducibility.

The available commit evidence was also insufficient to reconstruct a complete commit-by-commit development timeline. The report therefore does not claim that every repository file, commit, or external Drive artifact was exhaustively inspected.

3 Workspace Setup and Project Organization

The project was developed using Google Colab, Google Drive, and GitHub. Google Drive was used as persistent storage for the dataset and generated artifacts, while Colab provided the execution environment. GitHub was used for version control and collaboration.

The workspace root used in Colab was:

```
/content/drive/MyDrive/image-captioning-rag
```

The main workspace structure observed during the project was:

```
image-captioning-rag/
|-- 00_original_data/
| |-- train2017/
| |-- val2017/
| `-- annotations/
|-- 01_manifests/
|-- 01_splits/
|-- 02_processed_data/
|-- 03_embeddings/
|-- 04_indexes/
|-- 06_releases/
|-- artifacts/
`-- reports/
```

The original dataset directory was named `00_original_data`. This exact directory name matters because earlier path errors resulted from using a different spelling.

The project notebooks divide the workflow into distinct stages: workspace setup, dataset audit, splitting and statistics, preprocessing, embedding generation and indexing, quality control, and a later final-split implementation.

3.1 Execution environment

The recorded Colab environment included:

- Python 3.13.18
- Linux kernel 6.6
- NumPy 2.1.3
- Pandas 2.2.3

The final embedding run used CPU execution because a GPU was not available in the recorded run. The reported image batch size was 8 and the caption batch size was 32.

Colab runtimes are temporary. The project therefore relied on Google Drive for persistent storage and required the Drive mount and workspace initialization steps to be rerun after runtime reconnection.

4 Dataset Audit, Scaling, and Knowledge-Base Rescaling

4.1 Source dataset

The project uses MS-COCO 2017, with the following annotation files:

- `captions_train2017.json`
- `captions_val2017.json`

- `instances_train2017.json`
- `instances_val2017.json`

The recorded source annotation inventory was:

Table 1: MS-COCO 2017 source annotation inventory

Dataset element	**Recorded count**
Train image records	118,287
Validation image records	5,000
Train caption annotations	591,753
Validation caption annotations	25,014

4.2 Incomplete local extraction and restoration

The local `train2017` directory initially contained 61,705 JPG files, although the training annotation file described 118,287 image records. The audit identified 56,582 training image files missing from that local directory when compared with the full annotation inventory.

For the selected 10,000-image knowledge base, 4,908 selected image files were initially unavailable locally. The split was not supposed to be silently reduced to the number of images already present on disk. Instead, the selected IDs were retained and the missing selected files were restored.

The restoration procedure reported:

Table 2: Targeted restoration of selected knowledge-base images

Measure	**Count**
Selected images initially missing	4,908
Images restored	4,908
Restoration failures	0
Selected images accessible after restoration	10,000

The investigation encountered an HTTPS certificate hostname mismatch when accessing the COCO image host. The project then used a targeted archive-access approach to restore only the required images, rather than downloading and extracting the entire training archive again. A sample restored file was checked and recognized as a valid JPEG.

4.3 Development sample versus final scale

The project used a 1,000-image sample during development and debugging. That sample was useful for testing preprocessing, embedding generation, index construction, and early quality-control logic. It must not be confused with the intended final knowledge base.

The final-scale embedding notebook later reported 10,000 knowledge-base images. The development-scale quality-control notebook, however, still reported 1,000 images. This distinction is important when interpreting the evidence: successful development-scale checks do not independently establish the quality of the final-scale release.

5 End-to-End Pipeline and Implementation

5.1 Workspace initialization

The workspace setup notebook mounts Google Drive, defines the project root, and prepares the directory structure used by later notebooks. This stage is necessary to ensure that generated data and artifacts persist beyond an individual Colab runtime.

5.2 Dataset audit

The dataset audit stage examines annotation metadata, image availability, and dataset consistency. It compares annotation records with local image files and identifies missing files. The audit exposed the incomplete local training-image extraction described in Section 4.

The audit evidence also records the source dataset counts shown in Table 2. The restoration of the 4,908 selected files addressed the missing-image issue for the chosen knowledge-base subset, but it did not constitute a complete extraction of all 118,287 training images.

5.3 Split construction

The intended split design recorded in `01_splits/splits_v1.json` was:

Table 3: Intended split configuration in `splits_v1.json`

Split	**Images**	**Source**
Knowledge base	10,000	COCO train2017
Validation	100	COCO train2017
Test	100	COCO val2017

The split manifest recorded seed 42 and version 1.0.0. It also recorded that validation images and test reference captions must be excluded from the knowledge-base index.

The split validation output reported:

- 10,000 knowledge-base image IDs.
- 100 validation image IDs.
- 100 test image IDs.
- Zero duplicate IDs.

- Zero missing IDs in the corresponding source annotation records.
- Zero overlap between knowledge base and validation.
- Zero overlap between knowledge base and test.
- Zero overlap between validation and test.

The knowledge-base manifest was reported to contain 10,000 rows and the columns `image_id`, `file_name`, `image_path`, and `split`.

5.4 Preprocessing

The preprocessing notebook includes image loading logic based on PIL, conversion to RGB, and image checks on a development sample. An image-loading helper using EXIF orientation correction was present in the preprocessing stage. However, the embedding notebook later defined its own image-loading helper that converted images to RGB without applying EXIF transpose, thereby overriding the earlier helper in that notebook's execution context.

A complete final-scale preprocessing audit for all 10,000 images was not established by the available evidence. In particular, a fully persisted resized-image collection and a complete record of the final image resize/crop configuration were not independently verified. The CLIP processor applies its own model-specific image transformations during embedding generation, but the exact preprocessing configuration should be documented and pinned in the release metadata.

5.5 Embedding generation

The final embedding notebook reports the following models:

- Image encoder: `openai/clip-vit-base-patch32`.
- Caption encoder: `sentence-transformers/all-MiniLM-L6-v2`.

The reported dimensions and final counts were:

Table 4: Final-scale embedding outputs

Embedding type	**Vector dimension**	**Number of vectors**
Image embeddings (CLIP)	512	10,000
Caption embeddings (Sentence-Transformers)	384	50,030

The notebook reports normalized embeddings. The model identifiers, embedding dimensions, batch sizes, device, and vector counts were included in the generated metadata. The exact model revision or immutable checkpoint hash was not pinned in the evidence reviewed.

5.6 FAISS index construction

The project uses FAISS `IndexFlatIP` indexes. With normalized vectors, inner-product similarity is equivalent to cosine similarity.

The final notebook reports:

- Image index size: 10,000 vectors.
- Caption index size: 50,030 vectors.
- Image index dimension: 512.
- Caption index dimension: 384.

The notebook also saves mapping files that associate vector positions with image IDs and caption records. The image-ID array and mapping parquet files are important because an index alone does not preserve the semantic relationship between a vector position and its source image or caption.

An image retrieval smoke test queried the image index using the first knowledge-base image embedding. The same image was returned at the top with a similarity score close to 1.0. This is a useful index-construction check, but it is not a held-out evaluation and does not establish that self-retrieval is prevented for indexed images.

6 Drive Structure and Generated Artifacts

The project workspace separates original data, manifests, splits, processed data, embeddings, indexes, releases, and reports.

The final embedding notebook recorded the following release location:

```
06_releases/
‘-- phase1_final/
    |-- embeddings_and_index/
    ‘-- reports/
        ‘-- embeddings_and_index/
```

The reported final artifact set included:

```
embeddings_and_index/
|-- captions/
| ‘-- knowledge_base_captions.parquet
|-- clip/
| |-- kb_image_embeddings.npy
| ‘-- kb_image_ids.npy
|-- index/
| |-- clip_image_index.faiss
| |-- clip_image_index_mapping.parquet
| |-- sbert_caption_index.faiss
| ‘-- sbert_caption_index_mapping.parquet
|-- sbert/
```

```
| `-- kb_caption_embeddings.npy
`-- embedding_metadata.json
```

These paths and filenames were recorded by the notebook. The binary files were not all independently opened and revalidated during this report review. Accordingly, the existence and integrity of each file at the current Drive location remain subject to a final artifact-level verification.

The quality-control notebook inspected a different location:

```
artifacts/embeddings_and_index/
```

That directory contained development-scale artifacts with 1,000 image embeddings and 5,001 caption embeddings. The mismatch between this location and the final release directory is a major reason why the existing quality-control result cannot be treated as final release validation.

7 Statistical Exploration and Dataset Results

7.1 Caption inventory

The split-and-statistics notebook reported 50,030 captions for the knowledge base, 500 validation captions, and 500 test captions.

The knowledge-base caption distribution was:

Table 5: Caption count per knowledge-base image

Captions per image	**Number of images**
5	9,971
6	28
7	1

The knowledge base therefore does not contain exactly five captions for every image. The pipeline appears to preserve the available COCO caption annotations rather than enforcing a strict five-caption cap. This produces 50,030 captions rather than exactly 50,000.

7.2 Object statistics

The reported object statistics for the knowledge base were:

Table 6: Object-count statistics

Statistic	Value
Mean objects per image	7.3272
Standard deviation	approximately 7.489
Median	4
Minimum	0
Maximum	62

The notebook also generated a top-50 COCO object-class frequency output, an object-count/scene-complexity histogram, and a caption-length histogram.

7.3 Caption language statistics

The recorded caption statistics were:

Table 7: Caption text statistics

Statistic	Value
Average caption length	10.4605 words
Median caption length	10 words
Vocabulary size	13,892

The notebook also generated a part-of-speech distribution and a bar chart, as well as a table of frequent object co-occurrence pairs.

7.4 Limits of the statistical analysis

The available execution evidence supports object-class frequencies, object counts, caption lengths, vocabulary size, part-of-speech distribution, and a list of frequent object co-occurrence pairs.

It does not establish that the project completed a full analysis of spatial relationships, a visual co-occurrence matrix, or attributes and scene context such as colour, shape, indoor/outdoor setting, or day/night conditions. COCO instance annotations provide object categories and bounding boxes, but they do not supply a complete ground-truth attribute inventory for all such concepts. Any future relationships inferred from bounding-box geometry should be explicitly labelled as heuristics rather than annotation ground truth.

8 Requirement-by-Requirement Evaluation

The following table evaluates the principal Phase 1 requirements against the evidence available during the review.

Table 8: Phase 1 requirement evaluation

Requirement	**Status**	**Evidence and assessment**
Workspace setup and directory organization	**Partially verified**	Workspace setup notebook and Drive directory structure are present. Colab/Drive workflow was used.
Dataset source inventory	**Verified**	COCO train/validation image and caption annotation counts were recorded.
Audit of local image availability	**Verified**	The audit identified missing training files; 4,908 selected knowledge-base images were subsequently restored with zero reported failures.
Knowledge base contains 10,000 images	**Verified**	Final embedding notebook reports 10,000 image embeddings and an image index with 10,000 vectors.
Validation subset contains 100 images	**Partially verified**	The canonical split manifest reports 100 validation images from train2017. A later notebook implements a conflicting split.
Test subset contains 100 images	**Partially verified**	The canonical split manifest reports 100 test images from val2017. The written specification also contains a conflicting requirement of 1,000 test images.
No overlap between splits	**Verified**	The split validation output reports zero overlap among knowledge base, validation, and test IDs for the canonical split.
Exactly five captions per image	**Partially verified**	The pipeline preserves the COCO caption annotations and reports 50,030 KB captions. Twenty-nine images have more than five captions.
Preprocessing of all final images	**Partially verified**	Image loading and RGB conversion are implemented, but a complete final-scale preprocessing audit and exact transformation record were not verified.
Image embedding generation	**Verified**	Final notebook reports a $10{,}000 \times 512$ image embedding matrix.
Caption embedding generation	**Verified**	Final notebook reports a $50{,}030 \times 384$ caption embedding matrix.
FAISS index construction	**Verified**	Final notebook reports image and caption indexes with 10,000 and 50,030 vectors respectively.
Final-scale quality control	**Not completed**	The QC notebook output reviewed is development-scale: 1,000 images and 5,001 captions. It does not validate the final release directory.

Requirement	Status	Evidence and assessment
Test leakage prevention	**Partially verified**	Split-level ID disjointness is reported, but final index membership must be checked against the exact selected test IDs and final release artifacts.
Held-out image retrieval evaluation	**Not completed**	The recorded retrieval smoke test queries an image already present in the knowledge base. No complete held-out evaluation is evidenced.
Independent verification of final binary artifacts	**Unable to verify**	Final artifact paths were recorded, but the complete Drive release files were not independently inspected during this review.

9 Overall Re-evaluation, Risks, and Recommendations

9.1 Conflicting split implementations

Two different split implementations are present in the project.

1. The split recorded in `01_splits/splits_v1.json` and used by the final embedding notebook assigns 10,000 train2017 images to the knowledge base, 100 different train2017 images to validation, and 100 val2017 images to test.
2. The later `06_final_split.ipynb` notebook assigns 10,000 train2017 images to the knowledge base, then samples 200 val2017 images and divides them into 100 validation and 100 test images.

The second implementation therefore changes the validation source and creates a different split route. It also reports 50,018 knowledge-base captions, rather than the 50,030 captions reported by the canonical split and embedding pipeline.

These implementations must not both be treated as authoritative. The team should select one canonical split policy, regenerate or confirm all dependent manifests and caption files, and record the chosen version in the release metadata.

9.2 Test-set size inconsistency in the specification

The Phase 1 specification contains an internal inconsistency:

- Sections 4.1 and 4.2 specify a test subset of 100 images and 500 captions.
- Section 4.4.2 specifies 1,000 test images.
- Section 4.4.3 contains the apparent typo “1,00 images”.

The implementation reviewed uses 100 test images, which is consistent with Sections 4.1 and 4.2, but not with the literal 1,000-image statement in Section 4.4.2. The team should obtain an explicit decision from the project supervisor or specification owner and correct the document before release approval.

9.3 Five-caption policy

The phrase "five captions per image" can mean either exactly five captions selected for every image or the original COCO caption annotations associated with each image. The current pipeline preserves all available annotations, resulting in 50,030 captions for 10,000 images. The reported distribution shows that 29 images have six or seven captions.

The team should explicitly choose one policy:

- preserve all source captions and document the variable count; or
- select exactly five captions per image using a deterministic and documented rule.

The choice must be applied consistently to the knowledge base, validation references, test references, manifests, and reported counts.

9.4 Leakage and self-retrieval

The canonical split validation reports zero ID overlap between the knowledge base, validation, and test subsets. This is positive evidence of split-level separation.

However, split disjointness alone is not sufficient to establish complete retrieval leakage prevention. The final image index must be checked directly against the exact test image IDs. The caption index must also be checked to ensure that no validation or test reference captions were indexed.

The image retrieval smoke test used an image embedding from the knowledge base to query the same knowledge-base image index. The image returned itself at the top with a score close to 1.0. This is expected when the query image is already indexed. It does not show that test images are included in the index, but it also does not demonstrate a self-exclusion mechanism.

For held-out test evaluation, the test image must not be in the knowledge-base image index. If the system later supports queries using images that are themselves indexed, an explicit self-match exclusion policy may be required, depending on the intended application.

9.5 Final-scale quality-control gap

The quality-control notebook currently inspects the development artifact directory and reports 1,000 image embeddings and 5,001 caption embeddings. The final embedding notebook reports 10,000 images and 50,030 captions in a separate release directory.

The quality-control notebook should be updated to consume the final release directory and should assert the expected final counts, vector dimensions, mapping lengths, finite values, index sizes, and split exclusions. A successful development-scale QC run must not be presented as final release validation.

9.6 Recommended actions before Phase 1 approval

1. Confirm the authoritative split policy and retire or clearly label the conflicting split implementation.
2. Resolve the 100-versus-1,000 test-image inconsistency in the specification.

3. Confirm whether the knowledge base must contain exactly five captions per image or preserve all COCO captions.
4. Regenerate or validate all manifests and caption artifacts from the selected canonical split.
5. Update the quality-control notebook to read from `06_releases/phase1_final/embeddings_and_index`.
6. Rerun final-scale QC for 10,000 images and the selected final caption count.
7. Check that the exact validation and test image IDs are absent from the image index and that their reference captions are absent from the caption index.
8. Verify the existence, sizes, checksums, and mappings of all final binary artifacts in Google Drive.
9. Record exact model revisions, preprocessing settings, software versions, and deterministic parameters in release metadata.
10. Freeze the final manifests, indexes, metadata, QC report, and release version together.

10 Evidence Appendix

10.1 Key numerical evidence

Table 9: Summary of principal reported counts

Measure	**Value**
COCO train2017 image records	118,287
COCO val2017 image records	5,000
COCO train caption annotations	591,753
COCO val caption annotations	25,014
Local train JPG files before targeted restoration	61,705
Selected KB images restored	4,908
Final KB images reported by embedding notebook	10,000
Final KB captions reported by embedding notebook	50,030
Final image embedding shape	$10{,}000 \times 512$
Final caption embedding shape	$50{,}030 \times 384$
Final image FAISS index size	10,000
Final caption FAISS index size	50,030
Development QC image count	1,000
Development QC caption count	5,001

10.2 Model and index configuration

Table 10: Reported embedding and indexing configuration

Component	Reported configuration
Image embedding model	`openai/clip-vit-base-patch32`
Image vector dimension	512
Image batch size	8
Caption embedding model	`sentence-transformers/all-MiniLM-L6-v2`
Caption vector dimension	384
Caption batch size	32
Execution device	CPU
Embedding normalization	Enabled / normalized vectors reported
FAISS index type	`IndexFlatIP`
Similarity interpretation	Inner product; equivalent to cosine similarity for normalized vectors

10.3 Final reported artifact paths

```
06_releases/phase1_final/embeddings_and_index/
    captions/knowledge_base_captions.parquet
    clip/kb_image_embeddings.npy
    clip/kb_image_ids.npy
    embedding_metadata.json
    index/clip_image_index.faiss
    index/clip_image_index_mapping.parquet
    index/sbert_caption_index.faiss
    index/sbert_caption_index_mapping.parquet
    sbert/kb_caption_embeddings.npy
```

10.4 Important interpretation notes

- The final embedding counts are supported by recorded notebook outputs; the final binary artifacts were not all independently inspected during this review.
- The 1,000-image / 5,001-caption quality-control outputs refer to the development artifact directory, not the final-scale release.
- The two split implementations produce different validation sources and different knowledge-base caption counts.
- The 100-image test implementation follows the 4.1/4.2 specification statements, while the 1,000-image statement in Section 4.4.2 remains unresolved.
- The project has demonstrated index construction and a basic retrieval smoke test, but not a complete held-out caption-generation evaluation.

11 Conclusion

Phase 1 has achieved substantial technical progress. The project workspace and dataset pipeline have been established, the selected 10,000-image knowledge base has been made accessible despite incomplete local extraction, and the final embedding notebook reports the generation of image and caption embeddings together with their FAISS indexes.

The statistical exploration provides an initial description of object counts, caption lengths, vocabulary, part-of-speech distribution, and object co-occurrence. The project also has a documented split manifest and evidence of zero overlap among the three intended subsets in that implementation.

Nevertheless, the project should be considered **technically implemented but not yet fully release-validated**. The conflicting split notebooks, inconsistent test-set size requirements, variable caption counts, and development-only quality-control outputs prevent an unconditional Phase 1 sign-off.

Once the canonical split and caption policy are approved, the final release artifacts are independently verified, and the quality-control notebook passes against the 10,000-image release, Phase 1 can be frozen with a reproducible and auditable set of manifests, embeddings, indexes, metadata, and validation results.